

Font Manipulation Demonstration

Based on arXiv:2505.16957 - Invisible Prompts, Visible Threats

1. The Technique

This technique modifies TrueType font glyph mappings using the formula:

$$\text{GlyphIndex} = \text{idDelta} + \text{Unicode}$$

By changing the idDelta value, we can make:

- Unicode character 'b' (0x0062) display the glyph of 'a'
- Visual appearance: 'a'
- Actual Unicode content: 'b'
- Copy-paste reveals: 'b'

2. Example Calculation

Given:

- Character 'b' has Unicode value: 0x0062 (98 decimal)
- Character 'a' has glyph at index: 68

To make 'b' appear as 'a':

- Calculate new idDelta = 68 - 98 = -30

Verification:

- $\text{GlyphIndex} = \text{idDelta} + \text{Unicode}$
- $\text{GlyphIndex} = -30 + 98 = 68$ ✓
- Result: Unicode 'b' now displays glyph of 'a'!

3. Text Transformation Example

Original Text (what you type):

hello world

Unicode values:

U+0068 U+0065 U+006C U+006C U+006F U+0020 U+0077 U+006F U+0072 U+006C U+0064

After Font Manipulation:

hello world

(Looks the same, but Unicode is different)

Research Paper: arXiv:2505.16957 - Invisible Prompts, Visible Threats

Actual Unicode values (after manipulation):
Generated by Font Manipulation Tool - For Educational and Research Purposes

U+0077 U+006F U+0072 U+006C U+0064 U+0020 U+0068 U+0065 U+006C U+006C U+006F